

Qwen 3.5 Plus

Q: I am a student who is a first-time learner of Python and its packages. Please provide an at-a-glance summary of how pandas has been used in the attached marimo notebooks (Wk02_Marimo_ZScore.py, Wk03_InteractivePlots_Plotly.py, Wk04_DataPreparation.py) to help me understand its usage better.

Based on the three Marimo notebooks you provided, here is an **at-a-glance summary** of how the `pandas` library is used. Think of `pandas` as the "Excel engine" inside Python—it handles tables of data (called **DataFrames**).

1. Loading and Saving Data

Before analyzing, you need to get data in and out of Python.

- **Read CSV:** Loads data from a file into a DataFrame.
 - **Code:** `df = pd.read_csv('public/sp500_raw_data_Backup.csv')`
 - **Seen in:** `Wk03` , `Wk04`
- **Save CSV:** Saves your processed DataFrame back to a file.
 - **Code:** `df.to_csv("public/sp500_ZScore_AvgCostofDebt.csv", index=False)`
 - **Seen in:** `Wk02` , `Wk04`

2. Inspecting Data

Once loaded, you need to see what you have.

- **Display:** Simply typing the DataFrame name shows a preview.
 - **Code:** `df`
 - **Seen in:** `Wk03` , `Wk04`
- **Unique Values:** Finds all distinct entries in a column (e.g., list of all Sectors).
 - **Code:** `df['Sector_Key'].unique()`
 - **Seen in:** `Wk03`
- **Display Options:** Temporarily changes how many rows print to the screen.
 - **Code:** `pd.option_context('display.max_rows', None)`
 - **Seen in:** `Wk03`

3. Cleaning Data

Real-world data is messy. Pandas helps fix missing or bad data.

- **Find Missing Values:** Checks which rows are empty (`NaN`).
 - *Code:* `df['Total_Debt'].isna()`
 - *Seen in:* `Wk03` , `Wk04`
- **Fill Missing Values:** Replaces empty spots with a number (like 0).
 - *Code:* `df['Total_Debt'] = df['Total_Debt'].fillna(0)`
 - *Seen in:* `Wk03` , `Wk04`
- **Drop Missing Values:** Removes rows that have empty spots in key columns.
 - *Code:* `df.dropna(subset=['AvgCost_of_Debt', 'Z_Score_lag'])`
 - *Seen in:* `Wk04`

4. Creating New Variables (Vectorization)

This is one of pandas' superpowers. You can do math on **entire columns** at once without writing loops.

- **Column Math:** Creates new columns using existing ones.
 - *Code:* `df['avg_debt'] = 0.5 * (df['Total_Debt'] + df['prev_debt'])`
 - *Seen in:* `Wk04`
- **Apply Functions:** Uses a custom function or `lambda` on every row.
 - *Code:* `df['Risk_Zone'] = df['Z_Score_lag'].apply(categorize_zscore)`
 - *Seen in:* `Wk03` , `Wk04`

5. Filtering and Sorting

Selecting specific parts of the data you care about.

- **Filter Rows:** Selects only rows that match a condition (like a filter in Excel).
 - *Code:* `df[df['Sector_Key'] == 'financial-services']`
 - *Seen in:* `Wk03` , `Wk04`
- **Sort:** Orders the data by a specific column.
 - *Code:* `df.sort_values(['Ticker', 'Year'])`
 - *Seen in:* `Wk04`

6. Advanced Time-Series Logic

Handling data that changes over time (like stock data per year).

- **Group By:** Performs operations separately for each company (Ticker).

- **Code:** `df.groupby('Ticker')`
- **Seen in:** Wk04
- **Shift (Lag):** Gets the value from the **previous row** (e.g., last year's debt).
 - **Code:** `df['prev_debt'] = ... .shift(1)`
 - **Seen in:** Wk04
- **Contingency Table:** Creates a summary table counting occurrences (like a Pivot Table).
 - **Code:** `pd.crosstab(index=..., columns=...)`
 - **Seen in:** Wk04

7. Preparing for Visualization

Pandas DataFrames are the standard input for plotting libraries.

- **Plotting:** Passing the DataFrame directly to Plotly or Altair.
 - **Code:** `px.violin(df, x="Year", y="Total_Debt")`
 - **Seen in:** Wk03 , Wk04
- **Manual DF Creation:** Making small tables just for chart elements (like reference lines).
 - **Code:** `pd.DataFrame({'Score': [z_score], 'Label': [ticker]})`
 - **Seen in:** Wk02

Key Concept for Beginners: "The DataFrame"

Across all three files, notice that `df` is almost always a **DataFrame**.

- **Think of it as:** A smart spreadsheet table.
- **Columns:** Are called **Series** (e.g., `df['Total_Debt']`).
- **Operations:** When you add two Series together, pandas matches them row-by-row automatically (Vectorization).

Progression Across Files

- **Wk02 :** Uses pandas lightly, mostly for saving fetched data and preparing small tables for charts.
- **Wk03 :** Focuses on **loading, inspecting, and filtering** data to make interactive plots.
- **Wk04 :** The most advanced usage. Focuses on **constructing new variables**, handling **time-series lags** (`shift`), and **statistical analysis** (`crosstab` , regression prep).